

Labour market training in Norway—effect on earnings

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Abstract

By means of a 6-year panel of annual earnings data, we estimate the impact of a labour market training programme (LMT) targeted at unemployed adults. Unlike most nonexperimental studies, we use an internal comparison group of rejected applicants to measure the counterfactual outcome of trainees. For a subsample of applicants, rejections were randomised on a course-by-course basis. Following Heckman and Hotz [J. Am. Stat. Assoc. 84 (1989) 862] we test alternative estimators. All models report positive effects of being offered training, though the effects are not always significantly different from zero. The linear control estimator reports significantly positive effects of training. Analysing courses with ordinary assignment, the pre-training test indicates a positive selection bias. Turning to applicants for courses with random assignment, the linear control model is not rejected by the pre-training test. The fixed effect, random growth and modified random growth models are rejected by the model restriction test; this is the case for courses with ordinary assignment as well as random assignment. We illustrate the value of an internal comparison group in estimating the counterfactual outcome of participants, by comparing the post-training outcomes of rejected applicants and unemployed nonapplicants. Rejected applicants have significantly higher post-training earnings than eligible nonapplicants, which indicates that self-selection into the programme is influenced by unobservables. In general, the results clearly support the view that reliable studies of training effects should be based on internal comparison groups. Moreover, the assumptions motivating nonexperimental estimators should be explicitly tested. © 2002 Elsevier Science B.V. All rights reserved.

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1. Introduction

In many countries, active labour market programmes (ALMPs) play an important role in attempts to combat high and persistent unemployment. Public spending on ALMPs exceeds 1% of GDP in many industrialised countries and amounts to more than 2% in the Nordic countries. At the same time, politicians, bureaucrats and scholars often question whether such measures really do help the unemployed get back to work. Enhancing the effectiveness of ALMPs was also a main policy area identified by the OECD Jobs Strategy for helping to reduce structural unemployment (OECD, 1994, 1996).

The international literature on evaluation of labour market programmes is extensive and growing. In particular, there are numerous studies from the United States; see Barnow (1987) for a summing up of the results from evaluations of the Comprehensive Employment and Training Act (CETA) programme, Bloom et al. (1997) for an experimental evaluation of the Job Training Partnership Act (JTPA) programme, and LaLonde (1995) on public-sponsored training programmes in general. Recently, there have been several European studies as well: Lechner (1999), Eichler and Lechner (2002) and Fitzenberger and Prey (1997) evaluate training programmes in East Germany. Zweimüller and Winter-Ebmer (1996) and Jensen et al. (1993) evaluate training programmes in Austria and Denmark, respectively. Breen (1991) evaluates a temporary employment scheme in Ireland, while Edin (1988), Björklund (1990, 1991, 1993), Ackum (1991), Regnér (1997, 2002) and Larsson (2000) evaluate a number of different labour market measures in Sweden. An updated summary of both empirical strategies and evidence on the effects of training programmes can be found in Heckman et al. (1999).

The major challenge of any evaluation study is to handle the potential selection bias. The difference in post-programme outcomes between participants and the comparison group may reflect unobserved factors, rather than a *causal* effect of the programme. Different nonexperimental methods that attempt to take account of selection have proven to produce estimates that are sensitive to model specification and to the assumptions involved. The discussion has been especially lively in the United States; see Burtless and Orr (1986), LaLonde (1986), Fraker and Maynard (1987), Heckman and Smith (1993, 1999), Ham and LaLonde (1996), Heckman et al. (1998), Dehejia and Wahba (1999) and Smith and Todd (in press). These questions are also discussed—and different tests of nonexperimental estimators suggested—by Barnow et al. (1980), Heckman and Robb (1986), Heckman and Hotz (1989) and Moffitt (1991).

The selectivity problem is also a main issue in the present evaluation of the Norwegian labour market training programme. The selectivity bias can be reduced by appropriate construction of the treatment and comparison groups, i.e., by comparing comparable people. In many nonexperimental analyses, members of the treatment group are compared with a random sample of nonparticipants with no connection at all to the programme to be evaluated. The advantages of using *internal* comparison groups in programme evaluations are discussed in, e.g., Bell et al. (1995). Heckman et al. (1997, Table 7) find evidence that program dropouts—persons who apply and are accepted into the program but do not

receive any services—do reasonably well as a comparison group using the JTPA experimental data as a benchmark.

In this study, we use an internal comparison group of eligible persons who *applied* for training, but were *rejected* by the administrators due to the limited capacity of the programme. This group is chosen to avoid bias related to self-selection *into* the programme when nonapplicants are used as a comparison group. We also avoid the self-selection *out of* the programme when no-shows and dropouts are used as a comparison group. The administrative selection due to the process of distributing a limited number of training slots is, however, still present. A random sample of the more widely used comparison group of unemployed nonapplicants is also available. We illustrate the importance of a relevant comparison group by comparing the post-training outcome for rejected applicants and unemployed nonapplicants.

The data at hand cover a large set of highly relevant personal characteristics, including information on pre-training labour market outcomes. This enables us to test the assumptions of the alternative models to be estimated, following Heckman and Hotz (1989). In addition, random assignment to training was implemented for a fraction of the courses within the programme. A comparison of the estimated effects of courses with ordinary assignment and courses with random assignment will indicate the importance of administrative selection bias. However, the implementation of the random assignment involved several problems of the type discussed by Heckman and Smith (1995), making a straightforward comparison of experimental and nonexperimental estimators difficult.

The empirical results of this study clearly illustrate the significance of the design of the treatment and comparison groups as means of reducing the problem of selection bias. Comparing applicants offered training and applicants rejected from training, all models report a positive impact on annual earnings, though the estimated impacts are not always statistically significantly different from zero.

We sort among alternative models using two types of specification tests: pre-program tests that check for a zero impact prior to training and model restriction tests that check for nonzero coefficients on variables that should have zero coefficients under the identifying assumptions for a particular model. Heckman and Hotz (1989) introduced such tests into this literature. Analysing courses with *ordinary assignment*, the pre-training test indicates a positive *selection bias*. For courses with *random assignment*, the linear control model is *not* rejected by the pre-training test. The fixed effect model (which corresponds to the difference-in-differences estimator), the random growth model (which corresponds to the second differences estimator) and modified random growth model are all rejected by the model restriction test for courses with ordinary assignment.

The importance of the comparison group issue is further illustrated by our analyses of the post-training outcomes of rejected applicants and unemployed nonapplicants. Rejected applicants have significantly higher post-training earnings than eligible nonapplicants (conditional on observed characteristics), which indicates that self-selection into the programme is influenced by unobservables.

In general, the results clearly support the view that reliable nonexperimental studies of training effects should be based on internal comparison groups. Moreover, the assumptions

motivating nonexperimental estimators should be explicitly tested. Our limited experience with experimental data indicates that the use of experimental data is a possible way out of the selection bias problem, but not always an easy way out.

The remainder of the paper is organised as follows. The institutional framework is described in Section 2. In Section 3, we present the design of the study and discuss indicators of success and treatment. Earnings profiles and other data are presented in Section 4. In Section 5, we present the econometric models and tests suggested by Heckman and Hotz (1989). Selection into training is discussed in Section 6. Next, Section 7 presents the results of estimating and testing different model specifications. The analyses in Section 8 illustrate the importance of having information on eligibility and motivation for training (self-selection) in order to construct a relevant comparison group. Conclusions are drawn in Section 9.

2. Institutional framework

In Norway, the government launched the first comprehensive plan of active labour market programmes (ALMPs) in the early 1980s. From 1981 to 1983, participation in programmes increased by 140%. The policy was revised in 1988 and, within a 2-year period, the number of participants in programmes increased by 400%. During the period covered by our data, 1988–1994, unemployment was high by Norwegian standards, with a peak in 1993. The rate of open unemployment increased from 3.2% of the labour force in 1988 to 5.5% in 1994.² The average number of persons engaged in ALMPs increased from 8000 in 1988 to 56,000 in 1994, which shows how ALMPs serve to dampen the labour market effects of business cycles. During the bottom of the slowdown in 1992–1993, almost 3% of the labour force participated in these programmes. As most ALMPs last for less than half a year, the total number of persons participating in programmes *during* 1 year is about twice the participation rate at a point in time.

The labour market training programme (LMT or *AMO-kurs* in Norwegian) is by far the largest programme, covering about 40% of all ALMP participants. The aim of LMT is to preserve and improve the skills of the unemployed and thereby to enhance their employability. The programme is organised as off-the-job courses, mainly targeted at unemployed adults. Moreover, a substantial number of people (re-)enter the labour market via the training programme. At the macro level, LMT may contribute to reduced mismatch between workers and employers, greater effective supply of labour and thereby reduced wage inflation, increased production and employment, as well as reduced unemployment. The net impact on the macro level also depends on any dead-weight loss, substitution and displacements effects; see Heckman et al. (1999) for a detailed discussion. The present study deals with the average direct effect for the participants only.

LMT may affect future employment and earnings for the participants through various channels. First, successful training helps trainees to accumulate human capital that is

² Open unemployment refers to persons registered as full-time unemployed and searching for a job.

relevant to potential employers. Increased human capital may have a positive effect on wages as well as the probability of employment. However, if training increases the reservation wage, this may have the opposite effect on the employment probability. Second, as training represents a meaningful activity to most participants, it may help to prevent social isolation and psychological problems during a period of nonemployment. This may in turn enhance job search efficiency and reduce the probability that unemployed workers drop out of the labour force. Third, LMT participation may represent a signal about unobserved characteristics like motivation and effort that are correlated with productivity. Potential employers may consider a personal unemployment record that includes LMT to be better than a record with only open unemployment. This “signalling effect” of LMT is crucially dependent on the reputation of the programme. Programmes associated with the long-term or low-qualified unemployed may give a negative signal to employers. Fourth, training has an opportunity cost as time available to ordinary job search activities is reduced. Various empirical studies show that labour programme participants have very low transition rates to ordinary employment during the programme period; see, e.g., Røed and Zhang (1999).

The Norwegian LMT programme is financed by the central government and organised by the local employment service under the supervision of the Directorate of Labour and the Ministry of Labour. The courses are provided by the employment service, often in co-operation with other public and private institutions. Vocational training dominates the course offerings, with a wide range of subjects and crafts covered. Most of the courses are short, from 5 to 20 weeks. The courses are free of charge. All participants get a training allowance. Those who are entitled to unemployment insurance benefits may opt to collect their benefits instead, as they exceed the allowance. Compared with other ALMPs such as employment schemes, the LMT courses are not very expensive. Figures from the Directorate of Labour indicate the monthly cost for a participant in 1991 was about NOK 5500 (the equivalent of 1994 NOK 5800 per month or 1994 US\$860).

LMT is available for all job seekers and participation is voluntary. However, for some courses, applicants must qualify through education, previous vocational training, or work experience in order to be eligible. The capacity of most courses is limited.³ The rate of rationing for each course depends on the number of qualified applicants and the capacity of the course.

The incentives to take part in ALMPs may depend on how programmes are related to the eligibility and exhaustion of unemployment benefits. The time spent in LMT and the allowances collected do not qualify the participant for unemployment insurance (UI). According to the Norwegian system, it is necessary to have earnings from an ordinary job in order to qualify for UI benefits.⁴ Unemployed persons who refuse to accept offers of training may lose their unemployment insurance benefit. This sanction is, however, rarely carried out.

³ Note that we describe how the program operated at the time of the study (i.e., in the early 1990s).

⁴ The Norwegian system differs from the Swedish one in this regard. However, until 1997 (i.e., in the period covered by our study), earnings received during a temporary employment program (but not during training) would help an individual to qualify for unemployment insurance benefits.

The ordinary assignment procedure for LMT goes as follows: for each course, there is a list of qualified applicants. The selection of whom to make an offer of training to is done by the local employment service and the programme administrator. Many different criteria are used, but few formal guidelines exist. The office is supposed to offer a labour market programme to persons for whom unemployment insurance benefits will soon expire;⁵ thus, this group is given priority. Other applicants with an unemployment record of more than six consecutive months are also often given priority. When asked for the selection criteria, representatives of local and regional employment offices give many different answers from which we quote: “Applicants with the best prospects are given priority”, “Honestly seeking and hard working unemployed are preferred”, “Those who will benefit most from training get it”, “Our first choice is unemployed with no other opportunities” and “We try to make a qualifying programme career for each individual”. These quotes illustrate the complexity of the selection process as well as the fact that modelling the selection into training can be difficult.

3. The design of the study

The analyses are based on data for 2616 persons registered as unemployed and *applying for LMT* in August or September 1991 in 6 of the 19 counties of Norway.⁶ The counties were selected for administrative purposes. In addition, we have data for a *random* sample of 2511 persons, registered as *unemployed* at the same time in the same six counties, *not applying* for LMT or other programmes.

To evaluate the impact of a programme like LMT, an experimental design where applicants are randomly assigned into a treatment group offered training and a control group not offered training seems attractive. It is commonly argued that such a design implies that the difference in the mean outcome for the two groups provides an unbiased estimate of the average impact of training. One-level randomisation does not, however, secure unbiased estimates in the presence of crossovers and dropouts (see the discussion in Heckman et al., 2000). Moreover, randomisation provides no precise answers regarding how the programme would function with normal assignment procedures or regarding the distribution of impacts within groups of participants. See Heckman and Smith (1995) for an assessment of this kind of social experiment. To acquire some of the benefits from an experimental design, it was decided to undertake a “small scale experiment” for this evaluation of LMT.

In three of the six counties, admission to 82 out of 191 courses was carried out by *random assignment on a course-by-course basis*; in what follows, these courses are classified as random assignment courses (R-courses). The local organisers were instructed to replace ordinary assignment procedures by a random process. All courses were advertised at the local employment offices as usual. The staff was told to follow the standard procedures with respect to the registration of all qualified applicants, including

⁵ In 1991, the maximum benefit period was 2×80 weeks (160 weeks); at present, it is 156 weeks.

⁶ We started out with a much larger sample of applicants; see the discussion in Appendices A.

the usual counseling, and not to screen the potential applicants. The applicants were informed that admission would be determined by a lottery. After the deadline for applications, applicants were numbered alphabetically. The local administrators received a randomly drawn course-specific list of numbers from the Directorate of Labour. *Offers of LMT* were given to persons according to the list. The list was also used for supplementary admission to cover for persons who turned down the offer before the training started. Thus, randomisation took place only at one level; it determined whether or not an applicant was offered training.

In preparing for the study, considerable effort was required to convince the staff at the local branches that randomisation was useful as a means to estimate the effects of LMT. To secure co-operation from the local organisers, some courses were exempted from random assignment.

These included courses in which co-operating employers insisted on normal assignment procedures, courses aimed at specific target groups, and follow-up courses. For such courses and for all courses in the other three counties, the standard procedures were used; in the following, these courses are classified as *ordinary assignment* courses (O-courses). Different locations as well as exclusions from the randomisation imply that the two groups of courses (R-courses and O-courses) have a different composition of course categories.

The regional branches reported that the randomisation procedures were implemented in most cases. However, the LMT enrolment procedures did give administrators an opportunity to circumvent randomisation. By “strategically” *inactivating* applicants, that is, by withdrawing their applications from the register before assigning the training slots, an administrator could ensure that only the applicants he preferred for a course were subject to the draw. There is some evidence that this may have occurred. A simple comparison of the courses with and without random selection shows that the average *rate of inactivation*, defined as number of inactive applicants to the number of total applicants, is much higher where randomisation was supposed to take place. See Torp et al. (1993) and Raaum et al. (1994) for more details.

Our experience from the experiment clearly demonstrates that random assignment is difficult to implement when local administrators are opposed to it. A successful experiment requires the involvement of motivated administrative bodies throughout the process, as well as control mechanisms that secure a correctly implemented random assignment procedure.

Almost identical courses are organised at different locations. Still, the numbers of applicants for specific kinds of training are too small to make *separate* analyses for each course. Instead, the course category that a person applied for is used as an applicant characteristic and included in the outcome equation in order to capture possible effects on the *level* of the labour market outcome.⁷

⁷ We also estimated two alternative specifications of the linear control model. First, course-specific impacts were estimated by interacting “offered training” with the eight different course categories. Second, we estimated separate effects for the aggregated course categories “Manufacturing”, “Public services” and “Office work”. In both cases, the average training effects are very similar to those reported in Table 2. Details are available from the authors.

The variation in random assignment ratios across courses implies that the simple “difference in means” estimator is not justified. The assignment ratio—the number of persons in the treatment group compared to the number in the experimental control group—differs between courses due to the variation in the supply and demand for slots. When different courses attract applicants with different post-training labour market prospects (earnings), due to personal characteristics, occupation or industry affiliation, the simple “difference in means” estimate will depend on the correlation between the random assignment ratio and earnings without training. Moreover, if course impacts are correlated with their assignment ratios, perhaps because more popular courses have higher impacts, then the mean difference in outcomes between the treatment and control groups will not provide a consistent estimate of the impact of the average course or for the average applicant.

In order to get a fairly homogenous group of applicants typical of most LMT courses, we excluded specific training programmes for immigrants and “miscellaneous courses” (i.e., short job search courses, rehabilitation courses targeted at specific groups, etc.). Moreover, all applicants for courses without capacity constraints (i.e., with no rejected applicants, whether random or not) are excluded from the analyses. If the most successful courses also are the most popular ones (relative to their capacity), the exclusion of nonrationed courses will lead us to overestimate the impact of all courses on all applicants who take them (though we may consistently estimate the impact for the courses actually in our sample).

3.1. *Treatment*

Not all applicants who are offered training actually complete the course they are offered. The decision whether to accept an offer or not and, later on, whether to complete the course or not, is often taken simultaneously with other decisions that are crucial for the probability of becoming employed. For example, due to a time lag from application to enrolment, persons who have already entered employment at the date of enrolment are likely to turn down the offer. If compliance—the propensity to complete given an offer—is nonrandom, a simple comparison of completing participants and rejected applicants will give a biased estimate of the “effect of completing training”. There are good reasons to believe that compliance is nonrandom as people may enter jobs, ordinary education, or other activities during the application period. On the other hand, less motivated applicants are also likely to be found among the noncompliers. To get an unbiased estimate of the impact of completing training, all the decisions mentioned have to be explicitly modelled. Rather than trying to model these decisions explicitly, we define the *treatment* of interest in this study to be “offered LMT”. This treatment parameter is often called the effect of the “intention to treat”. Thus, the comparison group without treatment consists of applicants rejected by administrative procedures (ordinary or random) on the same courses included in our treatment sample.⁸

⁸ All applicants intended to participate in training courses starting in August or September 1991.

Among those offered training, we find both persons who never started a course and persons who dropped out in the midst of one. Overall, about 75% of those offered LMT completed their course.⁹

In experimental training studies, rejected applicants are sometimes found to enter the same (or similar) programmes, maybe later or at other locations; see Heckman et al. (1999) for an overview of substitution and crossovers in U.S. studies. Since rejected applicants in our study were not excluded from training courses starting in other counties, or some months later in the same county, we feared that this kind of substitution would contaminate the comparison group. However, very few rejected applicants took part in training during the training period (August 1991–June 1992). The difference in the rate of participation between the treatment group and the comparison group is particularly large during the first 6 months of the training period, 80% versus 3% for courses with ordinary assignment and 75% versus 2% for courses with random assignment. It is also significant that in the following 6 months, the participation levels were 49% versus 7% and 48% versus 6% for O- and R-courses, respectively.¹⁰

To illustrate the value of an internal comparison group, we present additional results in Section 8 from an analysis based on an alternative design. In this design, unemployed nonapplicants are used to estimate the counterfactual mean outcome of the trainees. This design is commonly used in nonexperimental studies; see, e.g., Regnér (2002). In the analysis of Section 8, the effect we estimate is that of completing training, rather than of being offered training. The comparison group is drawn from the stock of unemployed nonapplicants at the time when the training started.

3.2. *Common support*

An important lesson to be drawn from the extensive analyses of U.S. training programmes by Heckman and various colleagues is that training effects should be estimated using data wherein the supports of the probability of selection (treatment) coincide across the treatment and comparison groups. See Heckman and Smith (1999), Heckman et al. (1998) or Heckman et al. (1999) for discussions. In this study, the common support condition is applied to the O-course applicants. Although a common support is largely provided by our use of an internal comparison group of rejected applicants, we exclude a small number of applicants outside the overlapping range of the predicted “intention to treat” probabilities. Details of the exclusion procedure appear in Section 6. We also apply the common support condition in the alternative design presented in Section 8, where course completers are compared with non-applicants.

⁹ The rate of completion is 79.5 and 74.4 for the O- and the R-courses, respectively. Within these two groups of courses, 10.7% and 5.1% dropped out in the midst of training, and 9.8% and 20.5% declined the offer of training.

¹⁰ Details available on request.

3.3. Success

According to the Norwegian labour market authorities, the main objective of LMT is to help the unemployed get stable jobs. Even if employment is the overall goal of the programme, several arguments favour the use of *post-training earnings* to measure programme effects.

The first argument is *relevance*. Post-training earnings in year t (y_t) can be decomposed into days of employment (e_t), average hours per day employed (h_t) and average wages per hour (w_t), which gives: $y_t = e_t h_t w_t$. Here, e_t measures how quickly the person enters employment as well as the stability of the job, while h_t depends on opportunities, qualifications and preferences of the individual. Part-time employment is common among LMT applicants, indicating that many are rationed with respect to working hours. If the training effect on earnings is due to longer daily working hours, this should be considered as a success in line with (re-)employment. The hourly wage reflects productivity and the quality of the employment match. If LMT contributes to more productive employees and better matches, these effects are obviously socially beneficial. As the Norwegian wage structure is fairly compressed—see, e.g., Barth and Zweimüller (1994) or Stewart (1996)—earnings mainly reflect the duration of employment. If there is a positive effect of training on earnings, we do not expect wage increases to be an important component.¹¹ The obvious drawback with annual earnings is that the estimated impact will not answer the question of *how* training generates earnings gains; see Ham and LaLonde (1996). As pointed out by LaLonde (1995), however, earnings gains most often result from higher post-programme employment rates.

Cost-benefit comparisons also favour using earnings as a measure to evaluate the programme effect. An efficiency evaluation that compares costs and benefits of the LMT programme requires a common measurement scale. As costs are measured in NOK, earnings effects represent the obvious starting point for estimating the social benefits of the programme. A complete cost-benefit analysis must of course also address the questions of dead-weight loss, substitution, displacement and so on, as well as program benefits other than increases in participant earnings.

Data quality often requires specific surveys to get measures that correspond with the outcome(s) of interest. On the other hand, data attrition and measurement errors often favour register data. If we restrict ourselves to information available from official registers in Norway, earnings data from tax registers are clearly more reliable than the available information on employment.

In line with the objective of LMT, earnings should include wages as well as income from self-employment. Transfers, unemployment benefits, social support and training

¹¹ Previous Norwegian evaluation studies typically focus on employment at a certain point in time in the post-training period. The effect of training on the *probability of employment* (based on survey data) is analysed in previous papers for the same sample of applicants used here (see, e.g., Torp et al., 1993; Raaum et al., 1995). These authors found that training increased employment in the short run (after 6 months), but did not find a significant effect after 12 months.

allowances ought to be excluded. Our data on earnings are collected from public tax and wage registers. Unemployment benefits are subtracted, implying that our earnings are very close to income from work, including the earnings of the self-employed.¹²

4. Covariates and earnings profiles

The distribution of individual characteristics appears in Table 1. Two aspects of the group averages should be kept in mind. First, different course categories attract different kinds of applicants. As the degree of rationing varies across courses, the reported means reflect differences within as well as between courses. Second, for O-courses and non-applicants, we report mean characteristics of the common support samples for which the predicted treatment probabilities lie in the same range.¹³

As far as pre-training activities are concerned, we see that applicants spent on average 5 months of the 30-month pre-training period as openly unemployed, 2 months on labour market training and another 2 months on other programmes. About 6% were out of the labour force in 1989 and 1990. Those offered training spent less time openly unemployed, but more time on labour market training, compared to those rejected for training. Nearly 50% of the applicants had more than 5 years of work experience, with no obvious difference between those offered courses and those rejected in this characteristic. About 10–15% indicated “other plans than job search after the course” when asked in the pre-training survey.

Comparing the characteristics of those offered training and those rejected by the ordinary assignment procedure, we find some modest differences. Applicants offered training are somewhat older, more educated and have longer work experience. The pre-training unemployment records of the two groups are fairly similar. The rate of LMT participation immediately *before* our training period starts is, however, somewhat higher for those offered training. Applicants offered training are less likely to have been out of the labour force in the pre-training years of 1989–1990 and less likely to have received the transitional allowance for lone parents with dependent children, compared to rejected applicants. At the time of application, those offered LMT are more likely to be entitled to unemployment insurance benefits. UI entitlement indicates previous employment of some length. Thus, according to our data, LMT is typically not offered to unemployed persons who have just entered the labour market. Rather, unemployed persons with some labour market experience seem to be given a higher priority by the programme administrators.

For the R-courses, we find some unexpected differences. The fraction of women as well as educational attainment is somewhat higher among those offered training. Foreign

¹² The available measure of income, “Income Qualifying for Pension” (PI), includes unemployment benefits and wage earnings from various labour market programmes. Training allowances are not included. For the whole period covered by our data, we are able to adjust the PI for unemployment benefits (estimated for 1990). Earnings from participation in labour market programmes other than training are difficult to sort out and thus remain included.

¹³ For O-course applicants, see Section 6. The construction of the common support sample where nonapplicants constitute the comparison group is explained in Section 8.

Table 1
Individual and course characteristics

Sample	Ordinary assignment		Random assignment		Nonapplicants (N= 2511)
	Offered (N= 1203)	Rejected (N= 642)	Offered (N= 511)	Rejected (N= 259)	
Age (years)	29.8487 (9.0946) [0.2622]	28.9704 (8.5233) [0.3364]	30.1250 (9.2012) [0.4066]	30.3900 (9.6837) [0.6017]	32.9192 (11.4575) [0.2286]
Women (fraction)	0.4638 [0.0144]	0.5810 [0.0195]	0.6172 [0.0215]	0.5251 [0.0311]	0.4273 [0.0099]
Citizenship (fraction)					
Norwegians	0.9501 [0.0063]	0.9081 [0.0114]	0.9609 [0.0086]	0.9305 [0.0158]	0.9538 [0.0042]
Other Scandinavians	0.0091 [0.0027]	0.0078 [0.0035]	0.0098 [0.0044]	0.0077 [0.0054]	0.0072 [0.0017]
Other Europeans	0.0100 [0.0029]	0.0093 [0.0038]	0.0078 [0.0040]	0.0270 [0.0101]	0.0155 [0.00259]
Non-Europeans	0.0308 [0.0059]	0.0748 [0.0104]	0.0215 [0.0064]	0.0347 [0.0114]	0.0235 [0.0030]
Education (fraction)					
Compulsory school	0.1097 [0.0090]	0.1464 [0.0140]	0.1035 [0.0135]	0.2046 [0.0251]	0.1959 [0.0079]
Upper secondary school—1 year	0.0906 [0.0083]	0.0997 [0.0118]	0.0762 [0.0117]	0.0736 [0.0162]	0.0685 [0.0050]
Upper secondary school—2 years	0.2552 [0.0126]	0.2726 [0.0176]	0.1875 [0.0173]	0.1351 [0.0213]	0.2780 [0.0089]
Upper secondary school—unknown	0.0865 [0.0081]	0.1121 [0.0125]	0.1484 [0.0157]	0.2201 [0.0258]	0.0430 [0.0040]
Upper secondary school—completed	0.3741 [0.0140]	0.2975 [0.0181]	0.3828 [0.0215]	0.2973 [0.0285]	0.2939 [0.0091]
Higher education	0.0840 [0.0080]	0.0717 [0.0102]	0.1016 [0.0134]	0.0695 [0.0158]	0.1159 [0.0064]
Work experience (fraction)					
Less than 1 year	0.1654 [0.0107]	0.1604 [0.0145]	0.1621 [0.0163]	0.1583 [0.0227]	0.0749 [0.0053]
1–2 years	0.1305 [0.0097]	0.1620 [0.0145]	0.1426 [0.0155]	0.1467 [0.0220]	0.0952 [0.0059]
3–5 years	0.1937 [0.0114]	0.2040 [0.0159]	0.2305 [0.0186]	0.2316 [0.0263]	0.1900 [0.0078]
6–10 years	0.1995 [0.0115]	0.2243 [0.0165]	0.1894 [0.0173]	0.1544 [0.0225]	0.2059 [0.0081]
More than 10 years	0.3109 [0.0134]	0.2492 [0.0171]	0.2754 [0.0198]	0.3089 [0.0288]	0.4341 [0.0099]
Course categories, administrative classification (fraction)					
Technical work	0.0623 [0.0079]	0.0389 [0.0076]	0.0293 [0.0075]	0.0386 [0.0120]	na
Public services, child care	0.1579 [0.0105]	0.3925 [0.0193]	0.2148 [0.0182]	0.2780 [0.0279]	na

Table 1 (continued)

Sample	Ordinary assignment		Random assignment		Nonapplicants (<i>N</i> =2511)
	Offered (<i>N</i> =1203)	Rejected (<i>N</i> =642)	Offered (<i>N</i> =511)	Rejected (<i>N</i> =259)	
Course categories, administrative classification (fraction)					
Administration	0.0407 [0.0057]	0.0452 [0.0082]	0.0859 [0.0124]	0.0695 [0.0158]	na
Office work, word processing	0.2452 [0.0124]	0.1308 [0.0133]	0.2734 [0.0197]	0.2664 [0.0275]	na
Other services	0.0973 [0.0085]	0.0950 [0.0116]	0.1836 [0.0171]	0.0541 [0.0141]	na
Building and construction	0.0391 [0.0056]	0.0093 [0.0038]	0.0469 [0.0094]	0.0656 [0.0154]	na
Offshore	0.0399 [0.0056]	0.0857 [0.0111]	–	–	na
Transport	0.0599 [0.0068]	0.0218 [0.0058]	0.0547 [0.0101]	0.1351 [0.1213]	na
Manufacturing	0.2577 [0.0126]	0.1807 [0.0152]	0.1113 [0.0139]	0.0927 (0.2931)	na
Rationing (applicants per 100 slots)	221.7545 (143.8917) [4.1486]	368.8130 (229.1997) [9.0458]	194.2935 (80.3225) [3.5498]	252.6352 (94.4799) [5.8701]	na
Labour market experience in the pre-training period (January 1989–June 1991)					
Open unemployment (number of months in January 1989–June 1991)	4.8221 (5.5352) [0.1596]	4.9346 (5.5197) [0.2178]	5.6289 (6.0970) [0.2695]	6.2471 (6.5805) [0.4089]	6.3477 (6.5174) [0.1301]
Unemployment benefit receipt (number of months in January 1989– June 1991)	3.7515 (5.4219) [0.1563]	3.3629 (5.3537) [0.2113]	3.8848 (5.6416) [0.2493]	4.3050 (6.1760) [0.3838]	5.0299 (5.9655) [0.1190]
LMT participation during 1990 (rate)	0.2527 [0.0125]	0.2523 [0.0172]	0.2383 [0.0188]	0.2471 [0.0269]	0.0964 [0.0059]
LMT participation during January– June 1991 (rate)	0.2352 [0.0122]	0.1978 [0.0157]	0.2441 [0.0190]	0.1776 [0.0238]	0.0538 [0.0045]
Other labour market programme participation during 1989– June 1991 (rate)	0.3225 [0.0135]	0.3240 [0.0185]	0.2578 [0.0194]	0.3359 [0.0294]	0.2047 [0.0081]
Out of labour force (0,1) January 1989– December 1990 (rate)	0.0585 [0.0068]	0.0685 [0.0100]	0.0605 [0.0106]	0.0772 [0.0166]	0.0422 [0.0040]
Benefits in the pre-training period (1989–1990)					
Unemployment benefits in 1989	3454 (12,692) [366]	3443 (12,128) [479]	2535 (10,334) [457]	4316 (14,383) [894]	3827 (13,571) [271]

(continued on next page)

Table 1 (continued)

Sample	Ordinary assignment		Random assignment		Nonapplicants (<i>N</i> = 2511)
	Offered (<i>N</i> = 1203)	Rejected (<i>N</i> = 642)	Offered (<i>N</i> = 511)	Rejected (<i>N</i> = 259)	
Benefits in the pre-training period (1989–1990)					
Unemployment benefits in 1990	11,117 (19,155) [552]	9955 (18,387) [726]	10,836 (19,323) [854]	12,481 (20,365) [1265]	16,107 (23,916) [477]
Transitional allowance for lone parents with dependent children in 1989 or 1990 (rate)	0.0449 [0.0060]	0.0717 [0.0102]	0.0625 [0.0107]	0.0540 [0.0141]	0.0366 [0.0038]
Entitlement, disability and plans at the time of application (summer 1991, rates)					
Entitled to unemployment benefits (0,1)	0.7348 [0.0127]	0.6900 [0.0183]	0.7246 [0.0198]	0.7066 [0.0283]	0.7467 [0.0087]
Other plans than to search for work (0,1)	0.1571 [0.0105]	0.1355 [0.0135]	0.1055 [0.0136]	0.1158 [0.0199]	na
Vocational disability	0.1097 [0.0090]	0.1044 [0.0121]	0.0859 [0.0124]	0.1083 [0.0193]	0.0462 [0.0042]
Annual earnings (NOK 1994)					
1989	79,127 (79,615) [2295]	65,631 (69,783) [2754]	74,032 (80,190) [3544]	67,046 (79,997) [4971]	110,506 (88,192) [1760]
1990	67,978 (72,858) [2295]	57,105 (62,788) [2478]	60,673 (70,740) [3126]	58,283 (78,127) [4855]	94,694 (85,124) [1699]
1991	48,609 (56,507) [1629]	55,129 (56,146) [2216]	38,699 (45,833) [2026]	45,992 (60,064) [3,7332]	59,700 (64,645) [1290]
1992	66,589 (65,005) [1874]	66,092 (63,296) [2498]	51,818 (54,800) [2422]	53,797 (65,113) [4046]	63,691 (70,227) [1401]
1993	94,715 (77,941) [2247]	77,986 (70,698) [2790]	76,638 (68,470) [3026]	65,747 (75,011) [4661]	77,245 (75,967) [1516]
1994	105,891 (77,983) [2248]	87,413 (72,699) [2869]	95,567 (72,124) [3187]	81,425 (74,434) [4625]	89,111 (81,600) [1628]

Applicants for labour market training by treatment status (offered/rejected) and assignment procedure (ordinary/random) and unemployed nonapplicants. Means, (standard deviations) and [standard errors of the mean].

The category *Other Scandinavians* includes people from Denmark, Finland, Iceland and Sweden. *Vocational disability* means that the applicant, according to the employment service register, was classified as “in need of vocational rehabilitation” due to limited work capacity, for physical or other reasons. *Transitional allowance for lone parents with dependent children* is a means-tested support for parents (mothers) while they (typically) improve their qualifications in schools, universities or various training programs. *Out of the labour force* means no earnings, unemployment or program participation during 1989–1990.

citizens are underrepresented among those offered training. In light of the within-course estimates of selection presented in Section 6 below, we attribute these differences to the heterogeneity of training courses. For example, applicants with low education are more likely to be found in courses with many applicants relative to the number of available training slots.

The last column of Table 1 reveals that nonapplicants are older and more experienced than applicants. Their unemployment record over the preceding 30 months is also somewhat different: nonapplicants spent relatively more time openly unemployed and relatively less time as programme participants, especially as participants in LMT and rehabilitation programmes.

Turning to earnings profiles, keep in mind that all of the groups are unemployed in the second or third quarter of 1991. Consequently, all of the groups experienced a fall in earnings in 1990 and 1991, as illustrated in Fig. 1.¹⁴ This is the so-called Ashenfelter's (1978) dip in earnings, a common feature of micro-econometric studies of this kind. Unlike many other nonexperimental studies, in our data, trainees and comparison group members experienced a similar drop in mean earnings. From 1992 on, mean earnings increase for all of the groups. Focusing on LMT applicants, we see that those offered training had higher pre-training earnings than those rejected, conditional on assignment procedure. This pre-training difference is considerably smaller for the R-courses than for the O-courses. Those offered training experienced a larger drop in 1991, which includes the training period. While rejected applicants had somewhat higher earnings in 1991, earnings in 1992 are more or less equal for the two groups (again conditional on assignment procedure). Note that a minority of the training participants was still in training during the first few months of 1992.

During the post-training period, 1993–1994, Fig. 1 shows that applicants offered training received higher earnings than rejected applicants, conditional on assignment procedure. The 1993, as well the 1994, earnings difference is larger for O-courses than for R-courses. Before–after earnings growth seems to be highest for those offered training. In contrast to the 1993 and 1994 differences in levels, the difference in earnings growth between those offered training and those rejected is highest for R-courses. Fig. 1 also shows that the average annual earnings of nonapplicants are higher in the pre-training period (1989–1990) and lower in the post-training period (1993–1994) compared to applicants for LMT.

Summary statistics from Fig. 1 and Table 1 indicate that training raises the post-programme earnings of the participants. The earnings profiles also suggest that the ordinary assignment procedure, on average, favours applicants with higher earnings capacity. Section 7 addresses the question of whether or not these patterns can be

¹⁴ Nominal earnings are deflated by an index of average wage growth in order to give a measure of earnings in real terms, i.e., relative to the average increase in earnings for all employees (1994 NOK). Deflating by the consumer price index only would imply that training effects were affected by average productivity growth. In Norway, where about 75% of employees have wages determined by collective bargaining, wage growth is largely determined by centralised negotiations. The estimated training effects should not depend on the outcome of a wage bargain that affects the earnings of all, or at least of most, employees.

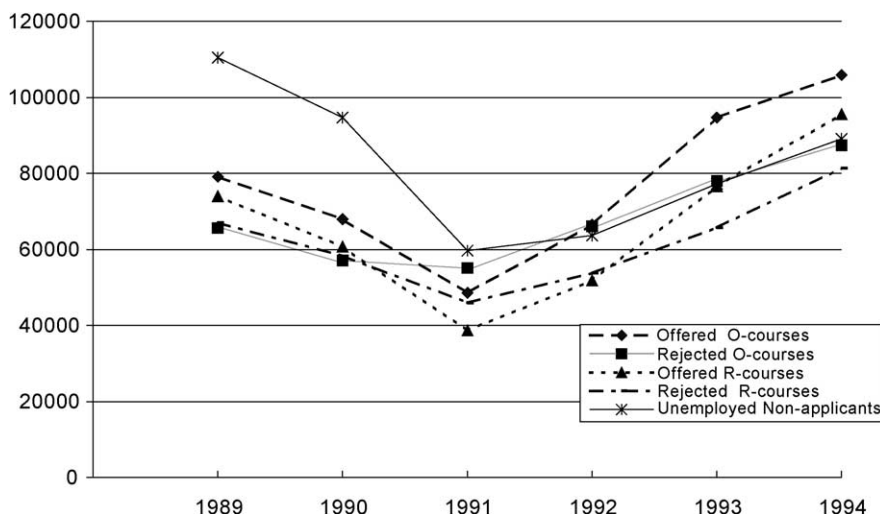


Fig. 1. Earning profiles 1989–1994 (1994 NOK). Applicants offered training, rejected applicants and unemployed nonapplicants.

explained by differences in observable characteristics. It is also evident that the earnings profile of the average nonapplicant differs considerably from that of the average applicant. This difference raises the question of using unemployed nonapplicants as a comparison group, an issue we elaborate on in Section 8. Before we report our empirical results, we now present the econometric framework suggested by Heckman and Hotz (1989).

5. Econometric models¹⁵

We have argued that annual earnings is a relevant outcome variable for evaluation of the training programme. Following most of the evaluation literature, we will look at earnings in *levels* rather than *logs*, which is the tradition within the wage determination literature.¹⁶

¹⁵ This section borrows heavily from the discussion in Heckman and Hotz (1989).

¹⁶ The standard empirical and theoretical arguments for log-linear models refer to a setting where earnings (or wages) represent a measure of “permanent” productivity (labour income). In our case, post-training earnings equal the product of hours worked per year and the average hourly wage. The major source of earnings variation is hours worked (duration of unemployment, stability of jobs, etc.). Consequently, zero is a legitimate outcome. The arguments in favour of log-linear models are therefore less relevant. A log-linear specification would also require that observations with zero earnings be given a positive value. When we estimate log-linear models, we find that the estimated training effects are highly sensitive to whether we replace the zero earnings observations with 1 NOK (about 1/8 US\$), 10 NOK or 100 NOK. Partly because of this sensitivity, we follow most of the evaluation literature in looking at earnings in levels rather than logs. However, if the true functional form is nonlinear, this choice may account for the poor performance of (some of) the models.

Let Y_{it}^1 be the earnings of individual i in period t if treated, and let Y_{it}^0 be the earnings for the same individual in the absence of treatment. In our setting, *treated* means being offered training, while *nontreated* means being rejected for training, in both cases conditional on being an applicant. The basic evaluation problem is that we do not observe each individual in both states. Let $D_i = 1$ for treated persons and $D_i = 0$ for nontreated persons. Then observed earnings Y_{it} are given by

$$Y_{it} = D_i Y_{it}^1 + (1 - D_i) Y_{it}^0.$$

Let α_t be the, by assumption, equal impact of training across individuals on earnings in period t . If training occurs in period k , then

$$Y_{it} = Y_{it}^0 + D_i \alpha_t \text{ for } t > k; \quad (1)$$

$$Y_{it} = Y_{it}^0 \text{ for } t < k.$$

We focus on the mean impact of treatment on the treated (in this case, the impact of being offered training on those offered it), defined by

$$\alpha_t = E(Y_{it}^1 - Y_{it}^0 \mid D_i = 1). \quad (2)$$

For post-programme periods,

$$E(Y_{it} \mid D_i = 1) - E(Y_{it} \mid D_i = 0) = \alpha_t + \{E(Y_{it}^0 \mid D_i = 1) - E(Y_{it}^0 \mid D_i = 0)\}. \quad (3)$$

Eq. (3) establishes that estimating the impact of training by simply calculating the mean earnings difference between the treated and the nontreated is valid if and only if expected earnings in the absence of training are the same for the two groups. By construction, an ideally implemented random assignment experiment will have this property. Selection bias exists when $E(Y_{it}^0 \mid D_i = 1) \neq E(Y_{it}^0 \mid D_i = 0)$. Various methods have been suggested to test for selection bias, and to correct for it when it is found. Our data allow us to apply a variety of such methods.

We confine our study to linear earnings equations, assuming that Y_{it}^0 is a linear function of observable characteristics, X_{it} , and unobserved characteristics, U_{it} . Thus,

$$Y_{it}^0 = X_{it} \beta + U_{it},$$

where β is a vector of parameters. We assume that $E(U_{it} \mid X_{it}) = 0$. The equations for observed earnings can be written as

$$Y_{it} = X_{it} \beta + D_i \alpha_t + U_{it}, \text{ for } t = 0, 1, \dots, T, \quad (4)$$

where $0, 1, \dots, T$ denote periods for which observations on Y and X are available.

The problem of selection bias arises when the treatment (the offer of training) is correlated with the unobserved characteristics of the individual. In such cases, $E(U_{it}|X_{it}, D_i) \neq 0$ and OLS will not give consistent estimates of α_t because $E(Y_{it}|X_{it}, D_i) \neq X_{it}\beta + D_i\alpha_t$.

A stochastic relationship between D_i and U_{it} may arise for a number of reasons. In the absence of random assignment, the behaviour of administrators and individuals determines whether or not each applicant gets offered training. In a nonhomogenous programme where different courses attract different sorts of people, the characteristics of the treated and nontreated populations will depend on how the degree of rationing is distributed across courses.

Whatever the decision-making procedure, we can describe it in terms of an index function framework, where the index, A_i , is a function of observed (Z_i) as well as unobserved (V_i) variables,

$$A_i = Z_i\gamma + V_i. \quad (5)$$

Treatment status is then given by $D_i = 1$ iff $A_i > 0$ and $D_i = 0$ otherwise.

The problem boils down to a potential correlation between U_{it} (unobserved variables) and D_i (treatment status). This dependence can arise for two, not mutually exclusive, reasons: (i) dependence between Z_i and U_{it} —“selection on observables” in the wording of Heckman and Robb (1986)—and (ii) dependence between V_i and U_{it} —what Heckman and Robb (1986) call “selection on unobservables”.

Selection on observables occurs when observed characteristics that influence selection into training *also* influence (post-programme) earnings. The main point is that by including a function of Z in the model, the expected value of the unobserved variables becomes independent of treatment status, i.e.,

$$E(U_{it} | D_i = 1, X_{it}, Z_i) = E(U_{it} | D_i = 0, X_{it}, Z_i) = E(U_{it} | X_{it}, Z_i).$$

In other words, as long as (X_{it}, Z_i) includes all of the variables that influence both A_i and Y_{it} , the selection bias disappears.

The *linear control function estimator* (LC) is a special case where $E(U_{it}|X_{it}, Z_i)$ is assumed to be linear. Under this assumption, we can reformulate the model and expand the explanatory variables to include Z_i , giving

$$Y_{it} = C_{it}\delta + D_i\alpha_t + U'_{it}, \quad (6)$$

where C_{it} denotes (X_{it}, Z_i) and $U'_{it} = U_{it} - E(U_{it}|C_{it})$ for both values of D_i ; see Barnow et al. (1980). In this case, the error term is now (conditionally) independent of the right-hand-side variable D_i , and a consistent estimate of α_t can be obtained by including all those variables that affect both selection and earnings in the absence of treatment in a standard OLS regression.

In the case of *selection on unobservables*, explicit assumptions about the time structure of the error term may solve the selection problem and thereby allow consistent estimates of

the treatment effects. But, of course, this is only the case if the assumptions correspond to the data generating process. Let U_{it} be decomposed as

$$U_{it} = \omega_{1i} + b_t \omega_{2i} + v_{it}, \quad (7)$$

where v_{it} is a zero-mean random component independent of all other values of $v_{it'}$ ($t > t'$) and of ω_{1i} and ω_{2i} . The dependence between U_{it} and D_i is assumed to emerge because of dependence between D_i and $(\omega_{1i}, \omega_{2i})$. Different estimators transform the dependent variable in various ways; that is, they impose different restrictions on the structure of Eq. (7).

The *fixed effect* (FE) estimator is based on the assumption that $b_t = 0$. Hence, the unobserved variables can be decomposed into a person-specific component, ω_{1i} , that is constant over time, and a random component v_{it} . Under this assumption, the expected value of the difference $U_{it} - U_{it'}$ is independent of U_i . Thus, $E(U_{it} - U_{it'} | X_{it}) = 0$ for all t, t' , where t (post-treatment) $> k$ (treatment) $> t'$ (pre-treatment). The fixed effect estimate is then derived from

$$\Delta Y_{it} = Y_{it} - Y_{it'} = (X_{it} - X_{it'})\beta + D_i \alpha_t + (v_{it} - v_{it'}) \quad (8)$$

for $t > k > t'$ where $(v_{it} - v_{it'})$ is the transformed random error term. The intuition behind the FE estimator is straightforward: by differencing the dependent variable, the fixed unobserved personal characteristics disappear. The FE estimator thus rests heavily on the assumptions that unobserved characteristics have the same impact on earnings in both the pre- and the post-treatment periods and that the selection problem springs solely from a correlation between the fixed effect and treatment.

The *random growth* (RG) estimator allows for person-specific earnings growth that is the same for all periods, i.e., $b_t = t$. Once again, we may transform the earnings equation in order to eliminate the person-specific unobserved characteristics. In this case, second differences yield

$$\begin{aligned} \Delta_2 Y_{it} &= (Y_{it} - Y_{it'}) - (t - t')(Y_{it'} - Y_{i,t'-1}) \\ &= D_i \alpha_t + \{(X_{it} - X_{it'}) - (t - t')(X_{it} - X_{i,t'-1})\}\beta \\ &\quad + \{(v_{it} - v_{it'}) - (t - t')(v_{it} - v_{i,t'-1})\}, \end{aligned} \quad (9)$$

where $t > k > t'$. The OLS estimator of α_t based on Eq. (9) is consistent under standard conditions, as all the v_{it} are white noise. The problem with the RG estimator in our context is the assumption that earnings grow at a *constant person-specific rate*. Typically, training applicants experience a fall in earnings the year(s) before the programme starts. However, we may extend the RG estimator to allow for person-specific growth that varies over time.

The *modified random growth* (MRG) estimator allows the growth component to be unrestricted. If $b_t \neq b_{t-1}$, we may write (after some manipulations),

$$\begin{aligned} \Delta Y_{it} = (Y_{it} - Y_{it'}) &= D_i \alpha_t + \tau(Y_{it'} - Y_{i,t'-1}) + (X_{it} - X_{it'})\beta \\ &\quad - \tau(X_{it'} - X_{i,t'-1})\beta + (v_{it} - v_{it'}) - \tau(v_{it'} - v_{i,t'-1}), \end{aligned} \quad (10)$$

where $\tau = (b_t - b_{t'}) / (b_{t'} - b_{t'-1})$ is a parameter to be estimated. (In the random growth model $\tau = t - t'$, i.e., the number of periods between the post-treatment and pre-treatment periods.) However, not all of the regressors are exogenous in Eq. (10) because $(Y_{it'} - Y_{i,t'-1})$ is correlated with $(v_{it'} - v_{i,t'-1})$. Thus, an instrument for $(Y_{it'} - Y_{i,t'-1})$ is necessary. As instruments, we need variables that do not enter the earnings equation. Such variables may, however, not be at hand.

We test the specifications in the various models above by means of two different methods. The *pre-training test* (PTT) utilises pre-treatment earnings and regressors for future participants and nonparticipants. The PTT tests test the joint null hypothesis that the correct model has been adopted for the post-treatment period *and* the over-identifying restriction that the same model holds in the pre-treatment period. The PTTs check whether an individual's treatment status enters the earnings equation in pre-treatment periods. If pre-treatment earnings are statistically significantly different for (future) participants and nonparticipants, it seems plausible that the source of this difference remains present with regard to post-treatment earnings. Hence, if the PPT fails, we are not convinced that an observed difference in post-treatment earnings can be interpreted as the impact of treatment, rather than reflecting a systematic difference in unobserved variables.

The LC estimator is rejected by the PTT if the estimate of $\alpha_{it'}$ from

$$Y_{it'} = C_{it'}\delta_{it'} + D_i\alpha_{it'} + U_{it'} \quad (6')$$

where $t' < k$ (training period) and $D_i = 1(0)$ for future treated(nontreated) persons is significantly different from zero. When lagged earnings enter $C_{it'}$, correctly implementing the PTT for the LC estimator requires access to pre-treatment earnings for multiple periods prior to the treatment period, but such information may not be available; see Section 7 for details. The structure of the PTT is similar for the FE, RG and MRG estimators of the earnings equation. If periods t , t' and $t' - 1$ all refer to pre-treatment periods, then training status should not significantly enter the earnings Eqs. (8), (9) and (10), respectively.

We can also test whether the model restrictions implied by each estimator hold empirically. Such *model restriction tests* (MRTs) cannot, however, be used to test the LC estimator, as the model underlying the estimator places no restrictions on what information should be included in $C_{it'}$. By contrast, the FE, RG and MRG estimators explicitly build on assumptions that imply that pre-treatment earnings from periods other than those specified in Eqs. (8), (9) and (10), respectively, should not appear as significant regressors when estimating these equations. Put differently, the MRT consists of simply adding an extraneous Y variable to the outcome equation. If the extraneous Y variable turns out to be significant, we reject the corresponding specification. For example, if the FE specification is correct, earnings in 1989 should not affect earnings growth from 1990 to 1993, or from 1990 to 1994.

There are limitations to this strategy; see, e.g., the discussion in Heckman et al. (1999). In particular, the PTTs cannot test the “just identified” version of the model where the relevant assumptions hold in the post-treatment period only. Moreover, the PTT has no power against selection on post-treatment earnings.

6. Selection into training and the common support

Selection into treatment among applicants (i.e., the probability of getting an offer of LMT) is estimated by logistic regression, with separate models for courses with random and ordinary assignment. Three different specifications, each with a different set of Z variables, are estimated and tested. Coefficient estimates for all three appear in Table A2 of Appendix B. Model III, the most restrictive model, includes only course characteristics and the county of residence. These variables capture supply side effects, as applicants living in areas with strong rationing of LMT or applying for a specific course with many applicants per slot are less likely to receive an offer. Model II extends the set of explanatory variables by including human capital variables and pre-training unemployment duration, as well as past participation in training and other labour market programmes. As explained in Sections 2 and 3, these human capital variables may be used as selection criteria by the local LMT administration. Model I includes all the variables in Model II along with pre-training earnings, unemployment benefits, benefit duration and a dummy for receiving a lone parent allowance during the pre-training period.

Empirically, the single most important variable determining whether or not an applicant receives an offer of LMT is the degree of rationing. Naturally, the larger the number of applicants relative to the number of slots, at the course level, the lower the probability of receiving an offer. As expected, the estimated effect of rationing is stronger for courses with random assignment than for courses with ordinary assignment.

Variables representing possible selection criteria, such as the applicant's position in the labour market as indicated by previous unemployment and programme participation, hardly affect the probability of treatment. This is as expected for courses with random assignment, but somewhat surprising for courses with ordinary assignment.

Testing the three models against each other by means of log-likelihood ratio tests gives the following results (exact test statistics appear in Table A2 of Appendix B). For courses with random assignment, the restricted model (Model III) is not rejected against the two others at the 5% level. Put differently, we fail to reject the restriction that personal characteristics do not affect the probability of an offer at conventional levels of significance. This suggests that random assignment was correctly implemented on a course-by-course basis, at least on those persons whose applications were not inactivated prior to random assignment. For courses with ordinary assignment, individual characteristics do matter, as the most restricted model (Model III) is rejected at the 1% level in favour of the model including personal characteristics (Model II). The estimates provide evidence of “creaming” as applicants with more pre-training unemployment and those not entitled to unemployment benefits have a lower probability of an offer. Adding pre-training income has, however, a very limited impact on the log-likelihood; thus, we fail to reject Model II against Model I.

6.1. Common support

We impose the common support condition using the estimated treatment probabilities from Model I in Table A2 of Appendix B. Figs. 2 and 3 display histograms of the densities

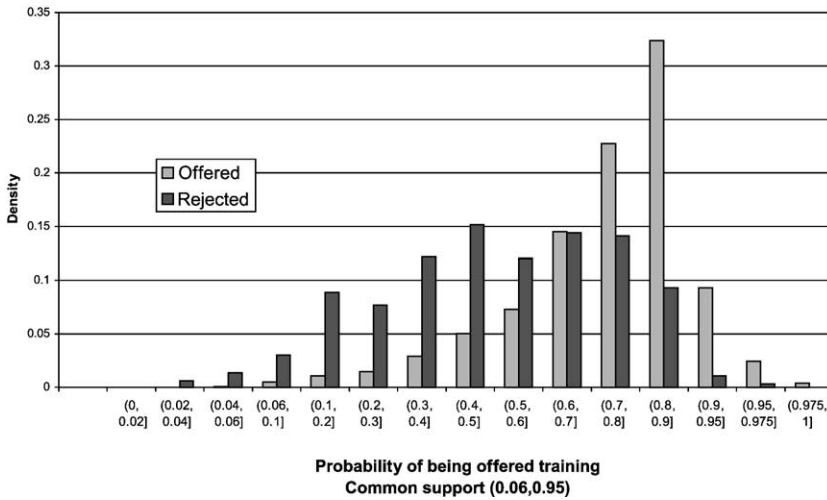


Fig. 2. Distribution of predicted probabilities: Prob(Offered training), O-courses.

of the estimated treatment probabilities for the O- and R-courses, respectively. Under random assignment, there is by definition no support problem and so we include all of the observations. Among the O-course applicants, we exclude applicants with estimated training offer probabilities below 0.06 and above 0.95, which represents 51 out of 1896 observations, or 2.5%. We define the common support region based on visual inspection of the distributions of the predicted probabilities rather than applying a formal procedure as in Heckman et al. (1997).

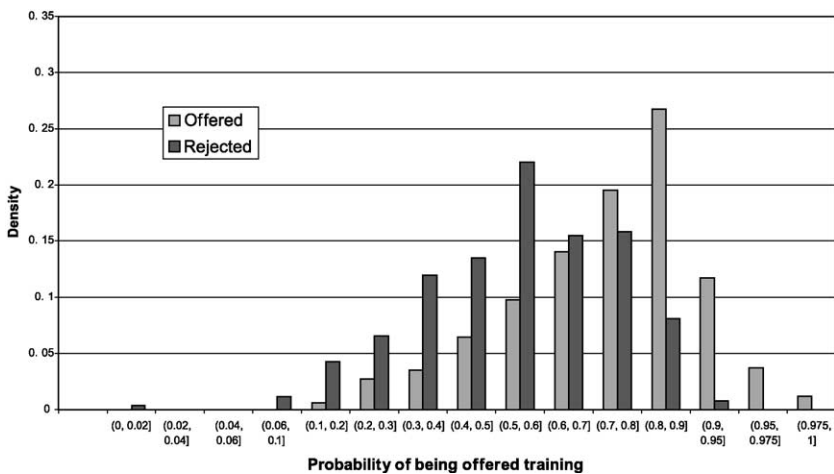


Fig. 3. Distribution of predicted probabilities: Prob(Offered training), R-courses.

7. Estimated training effects on earnings

All of the models we estimate yield positive training effects, but these effects are not always significantly different from zero at conventional levels. We find higher estimated earnings impacts in 1994 than in 1993. We also find larger estimated impacts for O-courses than for R-courses. None of the models based on “selection on unobservables” specifications pass the model restriction tests. The linear control model performs somewhat better. The pre-training test indicates, however, positive selection bias for the O-courses with the LC model. The following subsections review our findings in more detail.

7.1. *Linear control estimator*

The upper panel of Table 2 reports the estimated training effects from the linear control model for 1993 and 1994. We compare applicants offered versus rejected for O-courses in columns 2 and 5, and applicants offered versus rejected for R-courses in columns 3 and 6. The results of the PTTs appear in columns 8 and 9. Except for R-courses in 1993, all of the estimated training effects are significantly positive. The estimated effect for O-courses in 1993 is two times that for R-courses. For both groups, the training effect is higher in 1994, while the difference between the O- and R-courses is somewhat smaller in 1994.

The LC model includes a large number of controls. Table A3 of Appendix C shows how annual earnings in 1993 and 1994 correlate with the personal characteristics included in the LC model. We find that earnings increase with education, work experience and previous earnings. Women earn less than men, with the difference statistically significant in 1994. Conditional on actual work experience, earnings decrease with age, with the elderly having especially low earnings. Citizens from other countries have more or less the same earnings as Norwegians, conditional on other observed characteristics. The number of months as unemployed in the pre-training period has a strong negative impact on post-training earnings. Previous participation in a labour market programme for the disabled is also associated with significantly lower earnings. Finally, and not too surprisingly, persons with plans other than looking for a job had significantly lower earnings than job seekers.

Several course category dummies have large and highly significant coefficients, which illustrates the heterogeneity of the programmes offered and the resulting sorting into courses based on unobserved characteristics of the applicants. Applicants for courses in “manufacturing” serve as the reference group. Relative to this group, applicants for courses in “technical work” had higher earnings, while applicants for courses in “transportation, building and construction” had significantly lower earnings.

A comparison of Tables 1 and 2 reveals that the estimated impacts of training from the LC model are very similar to the unadjusted differences in mean earnings. This similarity appears despite the inclusion of a large number of control variables with significant effects on post-training earnings in the LC model. We believe that the similarity arises from the way in which we construct our comparison group using rejected applicants, rather than unemployed nonapplicants.

Table 2
Effects of training in 1991 on annual earnings in 1993 and 1994

	Impact on earnings 1993			Impact on earnings 1994			Pre-training test 1990		
	All (N = 2616)	Ordinary assignment (N = 1845)	Random assignment (N = 771)	All (N = 2616)	Ordinary assignment (N = 1845)	Random assignment (N = 771)	All (N = 2616)	Ordinary assignment (N = 1845)	Random assignment (N = 771)
<i>Model: Linear control (LC)</i>									
Offered training	16,916*** (2991)	19,926*** (3676)	10,088* (5373)	18,532*** (3088)	22,146*** (3802)	14,060** (5543)	5301** (2270)	9390*** (2790)	152 (4019)
Adjusted R ²	0.2172	0.2279	0.1961	0.1862	0.1901	0.1970	0.5068	0.4749	0.5729
<i>Model: Fixed effect (FE)</i>									
Offered training	7047** (3439)	6499 (4110)	8062 (6280)	9817*** (3596)	8948** (4298)	11,572* (6582)	– 2628 (2061)	– 1912 (2505)	– 4634 (3588)
Adjusted R ²	0.0116	0.0065	0.0165	0.0122	0.0106	0.0139	0.0086	0.0138	0.0002
MRT (1989)	– 0.4195*** (0.0200)	– 0.4023*** (0.0247)	– 0.4717*** (0.0337)	– 0.4632*** (0.0210)	– 0.4417*** (0.0260)	– 0.5105*** (0.0354)	na	na	na
MRT (1988)	– 0.2862*** (0.0195)	– 0.2524*** (0.0239)	– 0.3745*** (0.0333)	– 0.3338*** (0.0206)	– 0.3027*** (0.0254)	– 0.4013*** (0.0352)	na	na	na

Linear control (LC) and fixed effect (FE) estimates. Pre-training tests (PTT) and model restriction tests (MRT).

Estimated standard errors appear in parentheses. Level of significance: *** = 1%; ** = 5%; * = 10%. For the LC model, complete results for courses with random and ordinary assignment are presented in Table A3 of Appendix C. The FE model includes two covariates and the treatment indicator (offered training). The covariates are the change in the rate of unemployment in the applicant's home region and in the applicant's (previous) occupation from 1990 to 1993 or 1994. In the FE model, the left-hand-side variable in the PTT is the change in annual earnings from 1989 to 1990. The test variable in the MRT of the FE model is the level of annual earnings in 1988 and 1989. Complete results are available from the authors.

For the LC model, the pre-training test (PTT) checks whether training status enters the pre-training (1990) earnings equation, conditional on observables. An exact specification of the PTT would include earnings in 1987, and even 1986, because the LC model used to estimate the training effects contains earnings 3 to 5 years prior to training. Unfortunately, our data include only earnings in 1988 and 1989, and we cannot deduct unemployment benefits received in 1988 due to limitations in the registers. The autocorrelation in pre-training earnings implies that 1989 earnings capture most of the difference in 1990 earnings between the treated and the nontreated. As a result, the LC model is unlikely to be rejected if we include 1989 earnings as a control variable in the PTT. Therefore, in order to make the test as strong as possible, we include 1988 earnings instead, even though they contain unemployment benefits. For the O-courses, the PTT show significantly higher earnings in 1990 for those offered training. As observed characteristics alone do not account for the pre-training earnings difference, we conclude that there is positive selection into the O-courses. However, the estimated difference in 1990, conditional on control variables, is considerably lower than the estimated training effects for 1993 and 1994.

As expected, random assignment largely eliminates the selection bias problem. The PTT reveals that, conditional on observables, the pre-training earnings of those offered R-courses are very close indeed to the earnings of rejected applicants. We cannot, however, interpret the difference between the estimated impacts of the O- and R-courses as a direct estimate of the selection bias because the composition of the courses differs between the two assignment procedures.

The estimated training effects are large, about 15–20% of the average post-training earnings. They considerably exceed the percentage increase in earnings from an additional year of formal schooling in Norway, which is estimated by Barth and Røed (1999) and Raaum and Aabø (1999) to be about 7%. As the LMT typically lasts for less than 3 months, this informal specification test indicates that the point estimates of the training effect are too high. However, it is important to keep in mind that, unlike the standard marginal effects of schooling, our estimated training effects do not measure the impact on permanent earnings per unit of labour (hourly wages). Rather, the impacts on annual earnings in the first 2 or 3 years following LMT incorporate effects on both hours worked and wages per hour. The main factors behind earnings development in this transition period will likely be the time it takes before a person becomes employed and the stability of his or her new (and future) job(s). We expect the effects of LMT on permanent earnings to be considerably lower, as long-run differences between trainees and rejected applicants will consist primarily of any wage effects of training, rather than differences in hours worked due to an earlier return to work in the post-application period. In interpreting the estimated impacts, it is also important to keep in mind the width of the confidence intervals that surround them.

7.2. Fixed effect estimator

The FE model assumes that the person-specific component in the error term is constant over time, and that selection into treatment takes place based on this fixed component. This implies that the difference between earnings in the pre- and post-training periods results from training and from changes in individual and market characteristics during this

time. Our data contain no time-varying individual characteristics. The only explanatory variables that change from 1990 to 1993 are the two local unemployment rates we use: the overall unemployment rate in the applicant's home county (the county where LMT was arranged) and the rate in the applicant's previous occupation.

The lower part of Table 2 presents the results for the FE model for 1993 and 1994, with $(Y_{1993} - Y_{1990})$ and $(Y_{1994} - Y_{1990})$ as the dependent variables, respectively. The estimated impacts for 1993 are insignificant for both O- and R-courses. The estimated impacts for 1994 are, however, positive and significant at the 5% level. If we compare the FE results with the LC results for O-courses, we see that the estimated effects from the FE model are considerably lower. This finding supports the view that the linear control model overstates the effects of training for courses with ordinary assignment. For R-courses, the FE and LC estimates are very similar, as expected.

The PTTs indicate no significant correlation between pre-training earnings $(Y_{1990} - Y_{1989})$ and training. The MRT, however, clearly rejects the FE model. Changes in annual earnings from 1990 to 1993 and 1994 are highly correlated with earnings in 1989. The estimated coefficients on the lagged earnings variable in the PTT are significantly negative (around -0.4 to -0.5) in both 1993 and 1994 for both O- and R-courses. This indicates that a marginal increment in 1989 earnings of one NOK reduces earnings growth from 1990 to 1993 by about 0.4 to 0.5 NOK. The explanation for this negative correlation may be that persons entering the labour market in 1989–1990 from education or from unpaid work start out with zero or a low level of annual earnings and later experience relatively strong earnings growth. Whatever the explanation, the MRT indicates that the FE specification is wrong.

7.3. *Random growth estimator*

The random growth model extends the fixed effect model and allows for a person-specific annual earnings growth rate. By taking the earnings difference before and after training and subtracting a pre-training annual earnings change multiplied by the before–after time span, the person-specific components disappear. The upper panel of Table 3 shows that the RG model yields an insignificant estimated training effect for the O-courses. The estimated impact for the R-courses is somewhat larger and almost statistically significant.

In contrast to Heckman and Hotz (1989), who found the random growth model to perform relatively well, our MRT clearly rejects the RG model. Earnings in 1988 enter the RG equation with a significantly positive coefficient, while their coefficient should be zero under the assumptions that justify the RG model. To account for this rejection, recall that Fig. 1 reveals that all groups of applicants experienced a decline in mean earnings from 1989 to 1990. The extent of this drop is presumably strongly correlated with 1988 earnings at an individual level. Our set of control variables fails to explain this earnings dip; as a result, the data reject the assumption of time invariant earnings growth conditional on these variables.

7.4. *Modified random growth estimator*

As explained in Section 5, the MRG estimator allows for a person-specific earnings growth rate that varies over the sample period. There is one additional parameter,

Table 3
Effects of training in 1991 on annual earnings in 1993 and 1994

	Impact on earnings 1993			Impact on earnings 1994		
	All (N=2616)	Ordinary assignment (N=1845)	Random assignment (N=771)	All (N=2616)	Ordinary assignment (N=1845)	Random assignment (N=771)
<i>Model: Random growth (RG)</i>						
Offered training	15,457** (7576)	11,517 (9222)	22,303* (13,184)	20,106** (9539)	14,278 (11,605)	29,975* (16,509)
Adjusted R^2	0.0039	0.0070	0.0044	0.0088	0.0178	0.0018
MRT	0.3459*** (0.0436)	0.4212*** (0.0536)	0.2081*** (0.0745)	0.4956*** (0.0554)	0.5584*** (0.0682)	0.3709*** (0.0932)
<i>Model: Modified random growth (MRG)</i>						
Offered training	6505** (3452)	6045 (4127)	7713 (6283)	99,595*** (3602)	8740** (4299)	11,098* (6583)
τ	-0.5939*** (0.1021)	-0.6029*** (0.1205)	-0.4709*** (0.1772)	-0.6557*** (0.1066)	-0.6961*** (0.1257)	-0.4549** (0.1895)
Adjusted R^2	0.0258	0.0195	0.0257	0.0288	0.0302	0.0223
MRT	-0.2869*** (0.0188)	-0.2506*** (0.0231)	-0.3775*** (0.0319)	-0.3301*** (0.0199)	-0.2962*** (0.0245)	-0.4044*** (0.0338)

Random growth (RG) and modified random growth (MRG) estimates. Model restriction tests (MRT). Estimated standard errors appear in parentheses. Level of significance: ***=1%; **=5%; *=10%. The RG model includes the treatment indicator and two other covariates (as in the FE model). The MRG model incorporates six covariates, including the training indicator and an instrumented pre-training (1989 to 1990) change-in-earnings variable. As instruments, we use changes in labour market and unemployment status. The other covariates are changes in unemployment rates in the local region and in the previous occupation. For both models, we use annual earnings in 1988 as the test variable in the MRT. Analyses using annual earnings in 1989 as the test variable give similar results. Complete results are available from the authors.

$\tau = (b_t - b_{t'}) / (b_t - b_{t'} - 1)$, in the MRG model. The τ parameter equals zero under the assumptions of the FE model and equals three or four (in our context) under the assumptions of the RG model.

The modified random growth estimates of the training effect appear in the lower panel of Table 3. They are very different from those obtained by the RG estimator. However, the estimates of $\tau \approx -0.5$ clearly indicate that the RG model is a misspecification. The estimated impacts (including a value of τ close to zero) are in line with the FE estimates. As with the FE model, the model restriction test rejects the modified random growth model for all subsamples due to a correlation between the change in earnings before and after training and the pre-training earnings.

We use the 1989–1990 differences in full-time as well as part-time unemployment duration, along with spells of training and months of participation in two other labour market programmes, as instruments for $(Y_{1990} - Y_{1989})$ in the MRG model. These five variables explain about 10% of the variation in earnings growth from 1989 to 1990.¹⁷

¹⁷ No other variables are included and the R^2 value is 0.100 for all courses, 0.122 for R-courses and 0.097 for O-courses. In the training effect equation, the X variables include the regional and occupational unemployment rates.

Thus, there is no strong reason to believe that our results are biased due to weak instruments. On the other hand, we cannot be sure that our instruments are uncorrelated with earnings growth from 1990 to 1993 or 1994. As a result, the results from the MRG model should be interpreted cautiously.

7.5. Why do the “selection on unobservables” models fail?

Human capital theory models earnings as resulting from individual as well as contextual characteristics. If this relationship is stable, changes in earnings will only occur if some of these characteristics are altered. The FE, RG (and MRG) models are all based on the idea that the unobservable determinants of earnings have a certain time structure. Thus, the models are likely to be successful in cases where the earnings profile can be explained by the covariates at hand. Our data, however, include a very limited set of exogenous time-varying covariates. In results not reported here (but available from the authors), we estimate the various models separately for different subgroups of applicants—for men and women, for youths (18–25 years) and adults (above 25 years), for applicants with and without completed upper secondary education, as well as for applicants with different pre-training labour market histories. Splitting the sample into finer subgroups does not improve the performance of the “selection on unobservables” models. With few exceptions, the model restriction tests reject these specifications for the subgroups we examine.

7.6. Training effects by sex

Table 4 shows that we find some sex differences when we estimate separate models for men and women. We find larger training effects from O-courses for men than for women,

Table 4
Effects of training in 1991 on annual earnings in 1993 and 1994 (women and men)

	Impact on earnings 1993		Impact on earnings 1994		Pre-training test 1990	
	Ordinary assignment	Random assignment	Ordinary assignment	Random assignment	Ordinary assignment	Random assignment
Women	<i>N</i> = 931	<i>N</i> = 452	<i>N</i> = 931	<i>N</i> = 452	<i>N</i> = 931	<i>N</i> = 452
Offered training	17,879*** (4502)	14,453** (6358)	16,016*** (4662)	13,944** (6436)	7918** (3415)	4971 (4011)
Adjusted <i>R</i> ²	0.2420	0.1897	0.2070	0.1808	0.4883	0.4213
Men	<i>N</i> = 914	<i>N</i> = 319	<i>N</i> = 914	<i>N</i> = 319	<i>N</i> = 914	<i>N</i> = 319
Offered training	21,252*** (5981)	7011 (9652)	28,011*** (6203)	13,086 (10,334)	9314** (4565)	– 9178 (6988)
Adjusted <i>R</i> ²	0.2199	0.2161	0.1703	0.1665	0.4674	0.6402

Linear control estimates. Pre-training tests.

Estimated standard errors appear in parentheses. Level of significance: *** = 1%; ** = 5%; * = 10%. For a list of the covariates, see Table A3 of Appendix C. Complete results are available from the authors.

Table 5
Effects of training in 1991 on annual earnings in 1993 and 1994

	Impact on earnings 1993		Impact on earnings 1994		Pre-training test 1990	
	All (<i>N</i> = 3814)	Random assignment (<i>N</i> = 2881)	All (<i>N</i> = 3814)	Random assignment (<i>N</i> = 2881)	All (<i>N</i> = 3814)	Random assignment (<i>N</i> = 2881)
Completed training	17,939*** (2630)	8056* (4199)	20,691*** (2707)	18,054*** (4367)	− 6257*** (2172)	− 6615* (3689)
Adjusted <i>R</i> ²	0.1767	0.1874	0.2143	0.2398	0.4804	0.4762

Alternative design: completed training versus nonapplicants. Linear control (LC) estimates. Pre-training tests. Estimated standard errors appear in parentheses. Level of significance: *** = 1%; ** = 5%; * = 10%. Complete results for the “All” sample are included in Table A3 of Appendix C. The remaining results are available from the authors.

although the standard errors are too large to allow us to reject the null of equal training effects. The pre-training tests indicate positive selection bias for both men and women. For R-courses, women seem to gain more from training than men in 1993, but the sex difference is not present in 1994. The training effect for men is not significant, which may reflect the relatively small sample size.

8. Alternative design

Nonexperimental studies of labour market programmes typically use a sample of nonparticipants from the unemployment register to estimate the counterfactual outcome

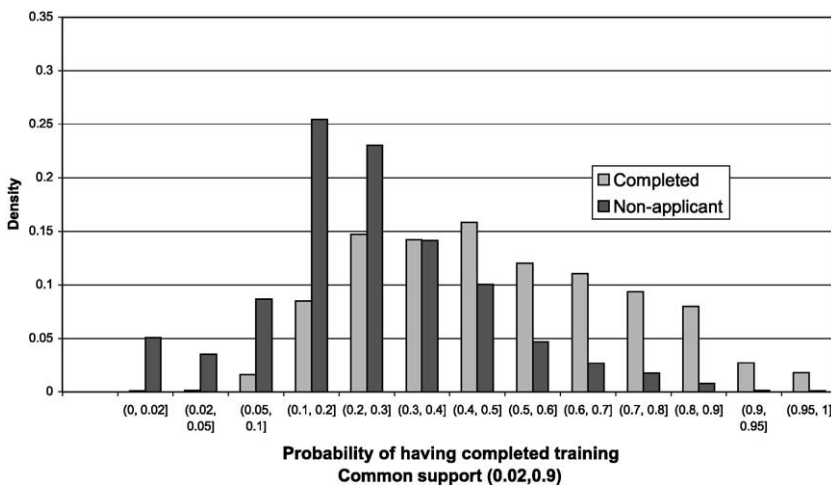


Fig. 4. Distribution of predicted probabilities: Prob(Completed training), all courses.

Table 6
Pre- and post-training tests

	Post-training test				Pre-training test	
	Earnings in 1993		Earnings in 1994		Earnings in 1990	
	All (<i>N</i> = 3899)	Random assignment (<i>N</i> = 3146)	All (<i>N</i> = 3899)	Random assignment (<i>N</i> = 3146)	All (<i>N</i> = 3899)	Random assignment (<i>N</i> = 3146)
Rejected applicant	7718*** (2887)	9448*** (3302)	8804*** (3044)	8580** (3441)	– 8469*** (2385)	– 7673*** (2387)
Adjusted <i>R</i> ²	0.1873	0.1896	0.2275	0.2306	0.4684	0.4634

Alternative design: rejected applicants versus nonapplicants. Difference in annual earnings in 1993 and 1994 (post-training test) and in 1990 (pre-training test). Linear control (LC) estimates.

Estimated standard errors appear in parentheses. Level of significance: *** = 1%; ** = 5%; * = 10%. Complete results are available from the authors.

of programme participants; see the overview in Heckman et al. (1999) and Regn r (2002). This section argues that such a design would have been inappropriate in our case.

Fig. 1 shows that the earnings profile of the unemployed nonapplicants is quite different from that of the LMT applicants. Although nonapplicants also experienced a drop in earnings in 1990 and 1991, their earnings fell from a considerably higher level. Earnings rose in the years 1992–1994, but at a much slower rate than for applicants. If data *had been* available for LMT participants and unemployed nonapplicants *only*, it seems likely that our analysis would conclude that training is extremely effective in raising the earnings of the participants.

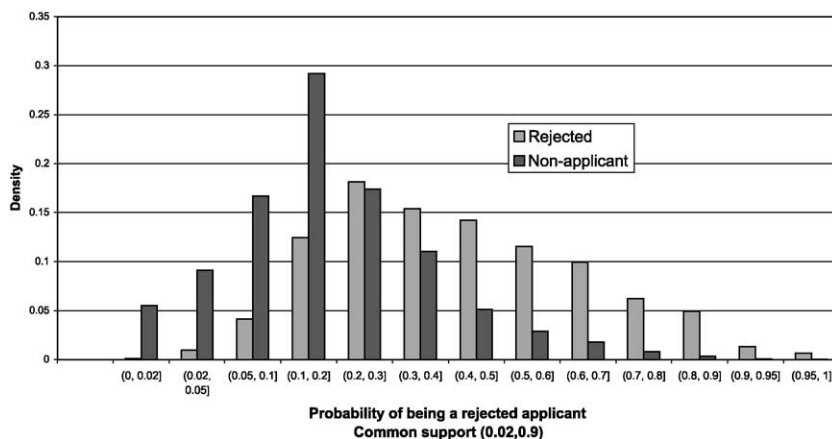


Fig. 5. Distribution of predicted probabilities: Prob(Rejected applicant), rejected and nonapplicants.

To test this conjecture, we estimate the LC model using applicants who completed training and a comparison group of unemployed nonapplicants. As in the main analysis using the rejected applicants, we impose a common support condition here as well, see Fig. 4.¹⁸ In Table 5, we report estimates for all courses (R- and O-courses combined) as well as for participants in R-courses only. The estimated earnings effects are statistically significant and somewhat higher than the corresponding estimates in Table 2. The exception is the effect of R-courses on earnings in 1993. As we did in estimating the effects of the offer of training in Section 7, we find somewhat larger effects in 1994 than in 1993. The PTT rejects the LC model for this comparison group. Surprisingly, the negative correlation between earnings in 1990 and subsequent training participation in 1991–1992 revealed by the PTT suggests that our estimated training effects are biased downwards. Thus, in the absence of data on rejected applicants, we would be led to conclude that the training effect was even stronger than the one estimated by the LC model in Table 5!

The post-training test in Table 6 that compares nonapplicants and rejected applicants demonstrates that this interpretation is wrong. Again we apply the common support condition. A histogram of the density of the estimated probabilities for the rejected applicants is presented in Fig. 5.¹⁹ Table 6 shows that rejected applicants had significantly higher earnings than nonapplicants in the post-training period, conditional on observables.¹⁹ Thus, all groups of applicants have higher post-training earnings than a random sample of unemployed nonapplicants.²⁰ Even those applicants rejected randomly had higher earnings than nonapplicants in the post-training period. This finding illustrates the value of information on eligibility and self-selection into training programmes in the construction of an appropriate comparison group. Without such data, even detailed information on pre-training labour market careers will likely not suffice to solve the problem of selection bias in estimating the impacts of training. In our context, applying the common support condition does not provide much assistance in avoiding biased estimates of the training effect. Our data indicate that training programmes attract applicants with better employment prospects than the average unemployed person. This kind of self-selection on post-training outcomes is difficult to identify and to correct for in the estimation.

¹⁸ The common support sample is defined on the basis of the predicted probabilities from a logit model with treatment (completing training in this case) as the dependent variable and observable characteristics available in the data for both applicants and nonapplicants as the independent variables. The covariates are the same as those in Model I in Table A2 of Appendix B, with the exception of the course-specific information, which is not relevant (and therefore not available) for the nonapplicants. A histogram of the density of the estimated treatment probabilities is included in Fig. 4. The region of common support is set to (0.02, 0.9). Almost 5% of the participant sample is excluded (62 of 1365), along with a bit more than 5% of the nonapplicant-sample (140 of 2651). Details are available from the authors.

¹⁹ The region of common support is set to (0.02, 0.9). The common support condition excludes 21 rejected applicants (15 from O-courses and six from R-courses out of a total of 657) and 149 nonapplicants.

²⁰ Conditional on a set of control variables that includes pre-training earnings.

9. Conclusions

The data we use to estimate the earnings effects of labour market training have a number of attractive features. First, we compare applicants offered training with rejected applicants. In this way, we avoid the selection bias related to selection into application. Second, the data include a large set of highly relevant personal characteristics. Third, the availability of pre-training earnings allows us to test the assumptions of alternative models. Fourth, random assignment—on a course-by-course basis—was implemented for some courses.

Looking at applicants for courses following the ordinary assignment procedures, we find that the linear control models report large and highly significant training effects on earnings. However, the pre-training test indicates selection on unobservables as pre-training earnings were higher for those offered training, even when estimating training effects separately for men and women. The pre-training earnings difference between applicants offered training and those rejected from training is only about half of the post-training difference. This suggests that not all of the post-training difference is due to more favourable unobserved characteristics among those offered LMT.

Our attempts to adjust for selection on unobservable components of earnings using fixed effect and random growth models were not successful. All of these models, with a few exceptions for smaller subgroups, were rejected by the model restriction tests. These results underscore the importance of specification tests, as, e.g., the widely used FE specification fails. Our selection-on-unobservables models presuppose a rather stable earnings profile from period to period, except for the training period. This implies that annual earnings in 1989–1990 and 1993–1994 should be part of such a stable pattern, except for variations explained by changes in covariates such as the rate of unemployment. The rejection of these models in our context indicates more complicated underlying earnings dynamics than those captured by these simple models.

Turning to the estimated training effects for the courses with random assignment, we find them to be smaller than those obtained for the O-courses. The estimated impacts are statistically significant at the 5% level in 1994 and at the 10% level in 1993.

Finally, to illustrate how the choice of comparison group may affect the estimated training effect, we present results from an analysis with a design frequently used in nonexperimental studies, in which we compare training participants with unemployed nonapplicants. The linear control estimate with this comparison group exaggerates the LMT effect. Moreover, the pre-training test gives the misleading impression that the effect is even higher. We show that self-selection is important, as rejected applicants have higher post-training earnings than unemployed nonapplicants. This result illustrates that information on eligibility and self-selection into a programme is extremely valuable for the construction of an appropriate comparison group to measure the counterfactual outcome of the treated.

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Appendix A. Gross and net samples

Due to administrative problems constructing the sample of LMT applicants (see Section 3) and a fairly low response rate in surveys (one pre-training and the two post-training), the final response rate is as low as 42%. Checking available register data on annual earnings, we find small differences between respondents and non-respondents. For the period from 1989 to 1993, Table A1 shows that average annual earnings are almost equal or slightly higher for the respondents ($N=5308$) compared with the original, gross sample ($N=12,733$).

The analyses of LMT applicants are based on a subsample of the respondents. Firstly, applicants for some specific courses are excluded, i.e., applicants for job search courses, rehabilitation courses and language courses for immigrants. Secondly, applicants for courses without capacity constraints (i.e., without rationing) are excluded. Thirdly, the matching procedure excludes a small number of persons from the sample of applicants for courses with ordinary assignment (51 persons).

Table A1. All LMT applicants. Gross sample and respondents. Annual earnings excluding unemployment benefits, 1989–1993 (1994 NOK)

Year	Gross sample ($N=12,733$)	Respondents ($N=5308$)
1989	75,575	77,154
1990	62,179	65,589
1991	47,344	47,547
1992	57,801	56,190
1993	73,257	74,138

Notes: The gross sample is the original sample, including all applicants for all courses in the selected counties with ordinary as well as random assignment. About 42% of the original sample answered the pre-training and post-training surveys. Nominal earnings are deflated by an index of average wage growth in order to give a measure of earnings in real terms, i.e., relative to the average increase in earnings for all employees (1994 NOK).

Appendix B. Selection

Table A2. The probability of getting an offer of training (treatment) among applicants for labour market training. Courses with ordinary and random assignment. Logistic regression, estimated parameters (estimated standard errors)

All covariates are dummy variables unless otherwise specified	Model I Ordinary assignment (<i>N</i> = 1845)	Random assignment (<i>N</i> = 771)	Model II Ordinary assignment (<i>N</i> = 1845)	Random assignment (<i>N</i> = 771)	Model III Ordinary assignment (<i>N</i> = 1845)	Random assignment (<i>N</i> = 771)
Intercept	1.5941*** (0.1983)	1.9958*** (0.5972)	1.6352*** (0.1941)	2.1605*** (0.5904)	1.2915*** (0.1364)	1.7613*** (0.5419)
Course categories (reference: manufacturing)						
Technical work	– 0.0560 (0.1749)	– 1.0042*** (0.3812)	– 0.0100 (0.1731)	– 1.0323** (0.3764)	0.0676 (0.1670)	– 0.6451* (0.3429)
Public services, child care	– 0.3244** (0.1594)	– 0.2977 (0.2792)	– 0.3336** (0.1584)	– 0.3599 (0.2744)	– 0.3400** (0.1429)	– 0.1068 (0.2401)
Administration	– 0.4700** (0.1925)	– 0.6606** (0.3041)	– 0.4528** (0.1907)	– 0.7019** (0.3001)	– 0.2970 (0.1811)	– 0.2667 (0.2670)
Office work, word processing	– 0.0030 (0.1314)	– 0.5949** (0.2562)	– 0.0151 (0.1305)	– 0.6316** (0.2527)	0.0854 (0.1171)	– 0.2987 (0.2213)
Other services	– 0.0931 (0.1579)	– 0.1597 (0.2984)	– 0.1201 (0.1575)	– 0.2105 (0.2941)	– 0.1780 (0.1478)	0.1366 (0.2605)
Building and construction	0.4476* (0.2566)	– 0.0470 (0.3190)	0.4109 (0.2552)	– 0.0838 (0.3168)	0.3106 (0.2496)	– 0.0704 (0.3060)
Offshore	0.1688 (0.2007)	– (0.1987)	0.1587 (0.1987)	– (0.1987)	0.2508 (0.1927)	– (0.1927)
Transport	0.3189 (0.1936)	0.2739 (0.3709)	0.2855 (0.1931)	0.32640 (0.3682)	0.2988 (0.1859)	0.0697 (0.3542)
Rate of rationing (applicants per 100 slots)	– 0.0024*** (0.0003)	– 0.0057*** (0.0009)	– 0.0024*** (0.0003)	– 0.0054*** (0.0009)	– 0.0023*** (0.0002)	– 0.0051*** (0.0008)
Pre-training earnings, benefits and labour market experience						
Annual earnings in 1990 (1000 NOK)	0.0004 (0.0008)	0.0032** (0.0014)	na	na	na	na
Annual earnings in 1989 (1000 NOK)	0.0022** (0.0009)	– 0.0024 (0.0015)	na	na	na	na
Unemployment benefits in 1990 (1000 NOK)	0.0036 (0.0031)	– 0.0019 (0.0049)	na	na	na	na
Unemployment benefits in 1989 (1000 NOK)	0.0007 (0.0030)	– 0.0063 (0.0046)	na	na	na	na

Open unemployment (number of months in January 1989–June 1991)	– 0.0083 (0.0108)	0.0053 (0.0144)	– 0.0108* (0.0065)	– 0.0068 (0.0088)	na	na
LMT participation during 1990	– 0.0622 (0.0874)	– 0.2105 (0.1386)	– 0.1191 (0.0852)	– 0.2322* (0.1343)	na	na
LMT participation during January–June 1991	0.0668 (0.0917)	0.2674* (0.1420)	0.0459 (0.0886)	0.2427* (0.1381)	na	na
Other labour market programme participation during 1989–June 1991 (rate)	– 0.0403 (0.0776)	– 0.1856 (0.1219)	– 0.0539 (0.0724)	– 0.2456* (0.1158)	na	na
Out of labour force during January 1989–Dec 1990	– 0.0543 (0.1545)	– 0.1759 (0.2337)	– 0.0933 (0.1500)	– 0.2694 (0.2245)	na	na
Entitled to unemployment benefits at the time of application (summer 1991)	0.0478 (0.1062)	0.1706 (0.1681)	0.2281*** (0.0873)	0.0479 (0.1335)	na	na
Other control variables						
Home county	Yes	Yes	Yes	Yes	Yes	Yes
Course characteristics, level of training	Yes	Yes	Yes	Yes	Yes	Yes
Human capital variables; sex, age, education, work experience, country of citizenship	Yes	Yes	Yes	Yes	No	No
Transitional allowance for lone parents and unemployment benefit duration	Yes	Yes	No	No	No	No
Model fitting information						
2 ln L (df)	1975.479 (48)	839.617 (45)	1989.412 (40)	838.832 (37)	2065.747 (18)	895.820 (15)
Concordant (percent of pairs)	77.3	75.4	77.0	74.6	74.5	69.0

Notes: Level of significance: *** = 1%; ** = 5%; * = 10%.

Appendix C. Linear control estimates

Table A3. Effects of training in 1991 on annual earnings 1993 and 1994 (1994 NOK). Linear control estimators (estimated standard errors). Complete results

All covariates are dummy variables unless otherwise specified	Ordinary assignment (N=1845)		Random assignment (N=771)		Alternative design (N=3814)	
	1993	1994	1993	1994	1993	1994
Intercept	52,555*** (11,542)	68,388*** (12,084)	83,123** (34,501)	90,833** (35,583)	60,120*** (7256)	77,948*** (7643)
Offer	19,926*** (3676)	22,146*** (3802)	10,088* (5373)	14,060*** (5543)	17,939*** (2630)	20,691*** (2707)
Woman	– 8103* (4448)	– 13,278*** (4567)	2682 (6674)	– 15,833** (6854)	– 13,196*** (2683)	– 23,859*** (2696)
Citizenship						
Other Scandinavians	– 20,788 (17,105)	– 5167 (17,703)	– 3799 (24,883)	– 6402 (25,676)	– 15,039 (12,359)	– 15,184 (12,726)
Other Europeans	18,221 (16,220)	4301 (16,781)	38,697* (20,084)	25,841 (20,725)	– 12,024 (9815)	– 10,551 (10,106)
Non-Europeans	2390 (8235)	– 4818 (8518)	1790 (15,371)	– 7571 (15,861)	– 8683 (7385)	– 21,645*** (7601)
Region and occupational characteristics						
County of Østfold	– 3631 (5964)	– 5883 (6171)	– 50,860* (29,330)	22,657 (30,269)	– 6518* (3899)	– 8959** (4014)
County of Hedmark	– 3681 (9651)	– 11,411 (9985)	– 30,228 (29,606)	56,374 (30,542)	– 6034 (4420)	– 7631* (4551)
County of Buskerud	– 1672 (5203)	– 10,582** (5383)	– 27,845 (48,324)	18,873 (49,862)	– 2005 (4136)	– 6802 (4258)
County of Vestfold	15,320*** (4974)	7818 (5146)	– 107,714 (71,406)	71,672 (73,678)	16,404*** (4305)	12,246*** (4434)
County of Sør-Trøndelag	10,030* (5999)	7546* (6207)	– 44,069 (29,373)	– 15,200 (30,299)	2442 (3863)	– 154 (3978)

Unemployment rate in previous occupation (1993)	– 1157* (610)	– 709 (727)	– 435 (941)	– 578 (1120)	– 586 (418)	– 301 (509)
Course characteristics (fields and level)						
Offshore	– 18,539* (9800)	– 4108 (10,142)	–	–	na	na
Technical training	23,904*** (8325)	20,696** (8604)	34,653* (17,806)	36,251** (18,324)	na	na
Public services, child care	15,029* (7715)	20,651*** (9803)	33,576** (13,765)	41,238*** (14,149)	na	na
Building and construction	– 15,112 (10,639)	– 19,850* (11,020)	– 31,085** (14,463)	– 21,819 (14,906)	na	na
Transport	– 13,946 (8706)	– 7299 (8990)	– 53,116*** (18,104)	– 52,405*** (18,681)	na	na
Office work, word processing	– 2851 (6184)	854 (6382)	273 (11,400)	1842 (11,734)	na	na
Other services	– 6824 (7601)	1548 (7844)	– 8554 (12,579)	– 10,164 (12,940)	na	na
Provides certificate	41,868*** (7056)	34,127*** (7299)	47,617** (19,800)	49,576*** (20,425)	na	na
Provides formal qualifications	1351 (4177)	709 (4324)	11,431 (6473)	6210 (6679)	na	na
Introductory course	1625 (5379)	12,939** (5565)	– 8498 (7754)	– 13,024 (8000)	na	na
Subsequent course	12,915 (13,657)	13,896 (14,128)	– 7422 (26,332)	– 30,539 (27,165)	na	na
Rationing (applicants per 100 slots)	13.66 (11.92)	16.43 (12.33)	34.93 (41.61)	0.67 (42.93)	na	na

Appendix C (continued)

Human capital variables						
Age (years)	– 334 (299)	– 532* (309)	– 910** (431)	– 1,121** (445)	– 580*** (181)	– 885*** (186)
Less than 19 years	– 6775 (7925)	3678 (8202)	– 23,270* (13,191)	– 9752 (13,603)	– 7272 (5905)	– 3673 (6079)
More than 60 years	– 49,120** (25,030)	– 55,042** (25,895)	47,647 (40,077)	– 76,241* (41,337)	– 41,261*** (10,768)	– 59,803*** (11,082)
Compulsory school	– 12,914** (5448)	– 16,538*** (5636)	– 1275 (8168)	– 7435 (8428)	– 16,841*** (3509)	– 19,612*** (3611)
Upper secondary school—1 year	– 12,750** (5834)	– 19,687*** (6088)	2240 (9770)	– 1911 (10,080)	– 13,672*** (4656)	– 19,764*** (4793)
Upper secondary school—2 years	– 9603** (4124)	– 7676* (4267)	7651 (7204)	5351 (7432)	– 9640*** (2952)	– 11,051*** (3039)
Upper secondary school—unknown	– 9701 (5977)	– 15,050*** (6183)	5541 (7426)	2616 (7661)	– 13,717*** (4981)	– 11,496** (5128)
Higher education	9256 (6373)	22,145*** (6585)	27,774*** (8878)	20,657** (9154)	13,758*** (4132)	19,069*** (4234)
Work experience	10,659* (6054)	6091 (6265)	12,572 (8892)	14,352 (9167)	18,399*** (5145)	15,272*** (5299)
1–2 years	17,431*** (6087)	18,183*** (6303)	8107 (8648)	7187 (8927)	15,521*** (4958)	16,369*** (5110)
3–5 years	22,288*** (6800)	24,140*** (7043)	18,037* (10,213)	22,940** (10,546)	21,156*** (5315)	25,417*** (5484)
6–10 years	17,543** (8147)	22,090*** (8436)	24,592** (11,968)	35,615*** (12,367)	26,018*** (6115)	29,062*** (6305)
more than 10 years						
Pre-training earnings, benefits and labour market experience						
Annual earnings in	0.091** (0.042)	0.029** (0.044)	0.220*** (0.067)	0.218*** (0.069)	0.161*** (0.028)	0.150*** (0.029)
1990 (NOK)						
Annual earnings in	0.136*** (0.038)	0.130*** (0.040)	0.047 (0.062)	– 0.043 (0.064)	0.077*** (0.027)	0.087*** (0.027)
1989 (NOK)						
Unemployment benefits in	– 0.348** (0.141)	– 0.320** (0.146)	0.256 (0.213)	– 0.694*** (0.220)	0.042 (0.082)	0.043 (0.085)
1990 (NOK)						
Unemployment benefits in	0.419*** (0.146)	0.094*** (0.152)	0.02 (0.23)	– 0.116 (0.240)	0.398*** (0.099)	0.304*** (0.102)
1989 (NOK)						

Transitional allowance for lone parents	– 13,087* (7259)	– 10,518 (7511)	– 27,368*** (10,448)	– 16,102* (10,777)	– 8368 (5951)	863 (6127)
Open unemployment (number of months in January 1989–June 1991)	– 961* (527)	– 1694*** (545)	– 1988*** (662)	– 2254*** (683)	– 1674*** (377)	– 1807*** (388)
Unemployment benefit duration (number of months in January 1989–June 1991)	185 (700)	966 (724)	1001 (952)	– 579 (982)	67 (471)	– 284 (485)
LMT participation during 1990	2948 (4216)	– 695 (4361)	– 4992 (6316)	– 211 (6520)	1777 (3617)	5592 (3724)
LMT participation during January–June 1991	– 7302* (4359)	– 6258 (4508)	1123 (6309)	3477 (6509)	2398 (4035)	3850 (4154)
Other labour market programme participation, 1989–June 1991	2444 (3716)	– 2070 (3844)	– 5874 (5653)	2603 (5832)	98 (2842)	– 3513 (2927)
Out of labour force, January 1989–December 1990	– 775 (7512)	– 7357 (7772)	– 11,394 (10,919)	– 8327 (11,266)	– 11,748** (5970)	– 16,138*** (6147)
Entitlement, disability and plans at the time of application (summer 1991)						
Entitled to unemployment benefits	14,066*** (5081)	11,310** (5257)	– 12,451 (7661)	232 (7904)	2237 (3450)	4388 (3552)
Other plans than to search for work	– 29,520*** (4526)	– 30,686*** (4683)	– 15,919** (7651)	– 19,002** (7972)	na	na
Vocational disability	– 27,951*** (5855)	– 24,761*** (6057)	– 36,690*** (9176)	– 33,957*** (9467)	– 16,201*** (4868)	– 20,287*** (5055)
Adjusted R^2	0.2279	0.1901	0.1961	0.1970	0.1767	0.2143

Note: The reference person is a Norwegian male, living in the county of Troms, applying for LMT in the field of manufacturing (not in the alternative design), 30 years old, who completed upper secondary school (3 years) and has less than 1 year of work experience. Annual earnings and unemployment benefits in 1990 and 1989 (NOK), months of open unemployment in the pre-training period and the level of unemployment in the previous occupation in 1993, are all measured from zero, which is also the value of these variables for the reference person. See Table 1 for further details.

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